Active Inference in Rust.... For Ants? (4-4-2024 Active InferAnts bundle of updates)

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all right it's April 4th 2024 this is an update from the active infer ants project where we're applying active inference models to ants in 2021 working with other authors we published an initial python code model ant simulation with Niche modification and forging with the single phermone code was written initially by Alec in that case adapting it from a previous publication with axle at all that was looking at cultural patterns of iates and with other nestmates in the subsequent years we continued to work on and explore these different ideas related to social physiology cyber physical systems multiscale planning and behavior and ant colonies and so on and today I wanted to bundle together a couple of updates related to the project as there's just a great Confluence of factors ranging from the RX and fur team's live stream on building a drone earlier today to just the inspiring work in the RX andur project at the instit and Holly's encouragement to look into rust so the big announcement SLT takeaway over at active inference Institute GitHub slactive infer ant let's start from scratch and multiple places and converge and diverge and have fun with ants and active inference and for the rest of this video I'm going to start zoomed in in the first section talking a little bit methodologically about how I used cursor. sh to translate RX and F Julia code and pmdp python code into new rust code and publish that open source using a method with cursor that pulls in GitHub repos and then we'll zoom out a little bit and look at how this draft SL sketch of the active inference package is laid out and what was retained through iterating locally on my side and as mentioned before I expect that we'll start from multiple places and multiple step by steps so I encourage that people get involved this fun project and today I'll just show a little bit that might spark some thoughts and uh let's just see how to also use these tools in a cool way so I'm starting in folder _ other with the Clone GitHub users python script this script does work not all the scripts work and the payload of this script is that for whichever GitHub usernames are provided it downloads all of the open repos that that user has into a folder called repo so here I've only left selected bias lab which is the developers of RX andur Julia and infer actively which is the hosting group of pmdp however I also curated some lists of a bunch of other active inference usernames and also some other things that are fun to bring in like bringing in energetics and quadres tetrahedral coordination geometries as well as a lot of ants and bioinformatics and also the exciting development in multi-agent Frameworks many could be added to all these lists but just to kind of show the minimal example what the current status of the repo does is it clones in the repos of bias lab and infer actively and puts it into other then by going to the gear on the top right of cursor and then features resync index cursor re indexes all of the code which helps it for when you do the AI chat you can do control enter and chat with the code base so that is a super helpful feature as is selecting text and control K or doing control shift L to move something over into the side window so all these are methods that I'm sure other people also have things that are exciting for them and there's probably many more improvements and meta improvements to come but this clone GitHub user script allowed me to bring in exactly the genres and the repos one by one in the repo script or in a batch download with the users and then cursor enabled me to chat with the codebase to explain it better and also to ask now write a pmdp styled script for this situation so um two ways in which I use that first I'll go through the rust read me and then some of the rust translations from the PDP and RX and fer and then we'll return to how the ants looked in the second section so this is in the systems design section of the repo because it's very preliminary the goals of today's work essentially where to use language models and code and document methods to translate and transform some open source Python and Julia active inference code into open source rust active inference code to develop a noisy non-expert draft sketch of an active inference ant colony boldy scale simulation in Rust language in order to learn highlight and explore the unique performance and security features of rust language for beginners such as myself and to provide an initial basis for further open source development at the intersection of active inference rust and ants so I used cursor 0.331 with the GPT form model I use the Clone GitHub user script to bring in as many repos as possible related to active inference or just a few resync the index in cursor copied over from the rest of the script some of the Python files and renamed them to be rust files and then here's some of the prompts that I used in the side chat and in the control K chat popup to help transform and iterate on improve the code so these were just a few of the prompts that helped where it got me was some Rust looking PDP style ants that I copied over and multiple separate attempts to write rust looking code not all of it was tested and a lot of it is still schematic but it's outlining at least an approach that with a rust expert in the loop I think we could have a great time with the generalized multiscale modeling of things thing class and this rust things starts in the kind of agent-based approach along the lines of the AOS active entity ontology for Science and then I also tried several other features based upon the code models know of rust so rust meta has some simulation environment aspects rust system check prints a lot of information about the computer rust meta configs has different kinds of affordances for the then configuration file which sets a bunch of the parameters which are used later and then in the rust worlds folder under config this is even more speculative so here we have cognitive security classes in Rust security classes in Rust and porting over from 4D Solutions quadres tetrahedral coordinates and Tetra volumes those were both translated from python so this just shows that by bringing in repos related to active inference we can transfer the semantics of examples and hopefully there will be a growing library of open- source examples in all these different languages and transfer and interpolate within and outside of those bounds to translate code and to get somewhere fast with natural conversation which is super exciting so now pulling back a little bit folder zero in the repo has some system specifications that were also iteratively written with language models and my put and then the folders are as follows nine other has some scripts related to cloning in repos six API is for API calls which is also just a kind of broadly outlined set of methods most of the simulating and preparing and all of the operating is in the scripts 1 through five so in one there's prepare so this has all of the agent-based definitions including pmdp ants that are defined and the general and the config set different configs most of these variables are not called however it's fun to explore and just build these hypothetical meta configuration files and then in the general preparation we have some scripts pointing towards aspects of cognitive security the course ordinates the IVM and the XYZ coordinates being able to flip between the coordination systems implementing some cryptographic metaphysical systems nesting networking Network science different physics including the metaphysics and the cognitive physics and then also for a little comedic relief the Salt Fat acid heat simulator in pmdp format so prepare gets all of the configurations ready for assembling the different things operate has a plan render and execute three-step process with a situational ant awareness Observer role that actually prepares plans out and then executes computationally the simulation measure has a set of scripts and methods related to specif ifying how you want to measure in a Quantum cognitive frame and then how you're going to summarize categorize and apply statistics to the measurements report is currently just for this release the specify report methods that detail how reporting is going to occur and then the specifications for followup which are the implications of the report so that's just one top level folder scheme for multi-layer simulations like the active inference package has to be and I look forward to hearing about some radically alternative starting points that we can also have in our Colony so hope this is at least interesting to some of you I encourage people with some or a lot or in between amount of familiarity with active inference and ants and all these different topics to come get involved and come to our weekly RX and fur meetings on Thursday join The Institute Discord or email blanket active inference do Institute or get in touch another way because this is going to be a fun project and really looking forward to seeing how we can all start over and collaborate on this further all right